02 BUS SELECT, GROUND CANCEL BUFFER, SUBGROUP

File: group_sheets/02_bus_select.kicad_sch

03 SUBGROUP

File: group_sheets/03_subgroup.kicad_sch

04 SUBGROUP, AUX MASTER

File: group_sheets/04_subgroup.kicad_sch

05 AUX MASTER

File: group_sheets/05_aux_master.kicad_sch

06 AUX MASTER, STEREO RETURN

File: group_sheets/06_aux_master.kicad_sch

07 STEREO RETURN

File: group_sheets/07_stereo_return.kicad_sch

08 STEREO RETURN, STEREO RETURN 2

File: group_sheets/08_stereo_return.kicad_sch

09 STEREO RETURN 2

File: group_sheets/09_stereo_return_2.kicad_sch

10 STEREO RETURN 2

File: group_sheets/10_stereo_return_2.kicad_sch

11 JACK BOARD LINK, JACK BOARD LINK 2

File: group_sheets/11_jack_board_link.kicad_sch

12 JACK LINK CLAMPS, POWER AND GROUNDS

File: group_sheets/12_jack_link_clamps.kicad_sch

13 POWER AND GROUNDS, RIBBON

File: group_sheets/13_power_and_grounds.kicad_sch

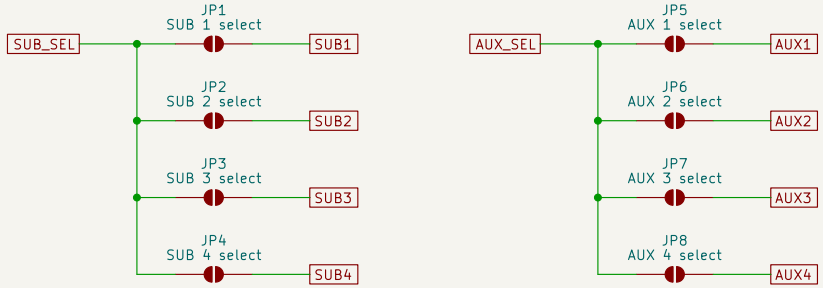
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)

Sheet: /
File: group.kicad_sch

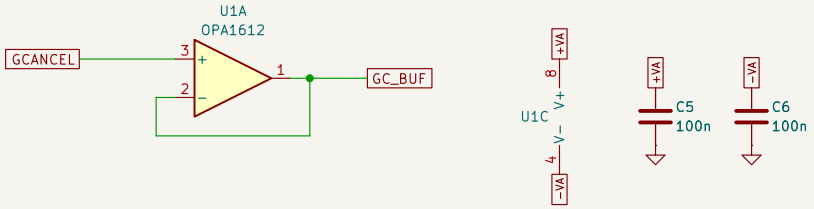
Title: Group module — goa console (sub + return + aux master)

Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 1/13

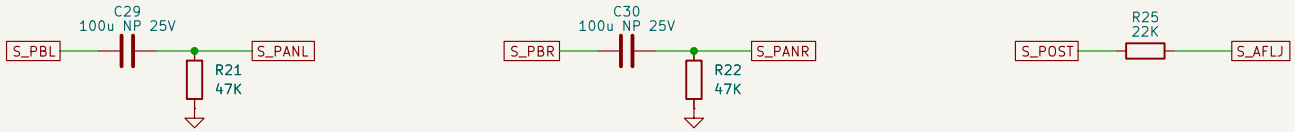
BUS SELECT close ONE SUB jumper and ONE AUX jumper: group n sums SUB n and AUX n (all open = summer idle)



GROUND CANCEL BUFFER (local, for this module's re-inverters; summing amps use the raw GCANCEL line)



SUBGROUP: sum (selected SUB bus), re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL [spec 12]



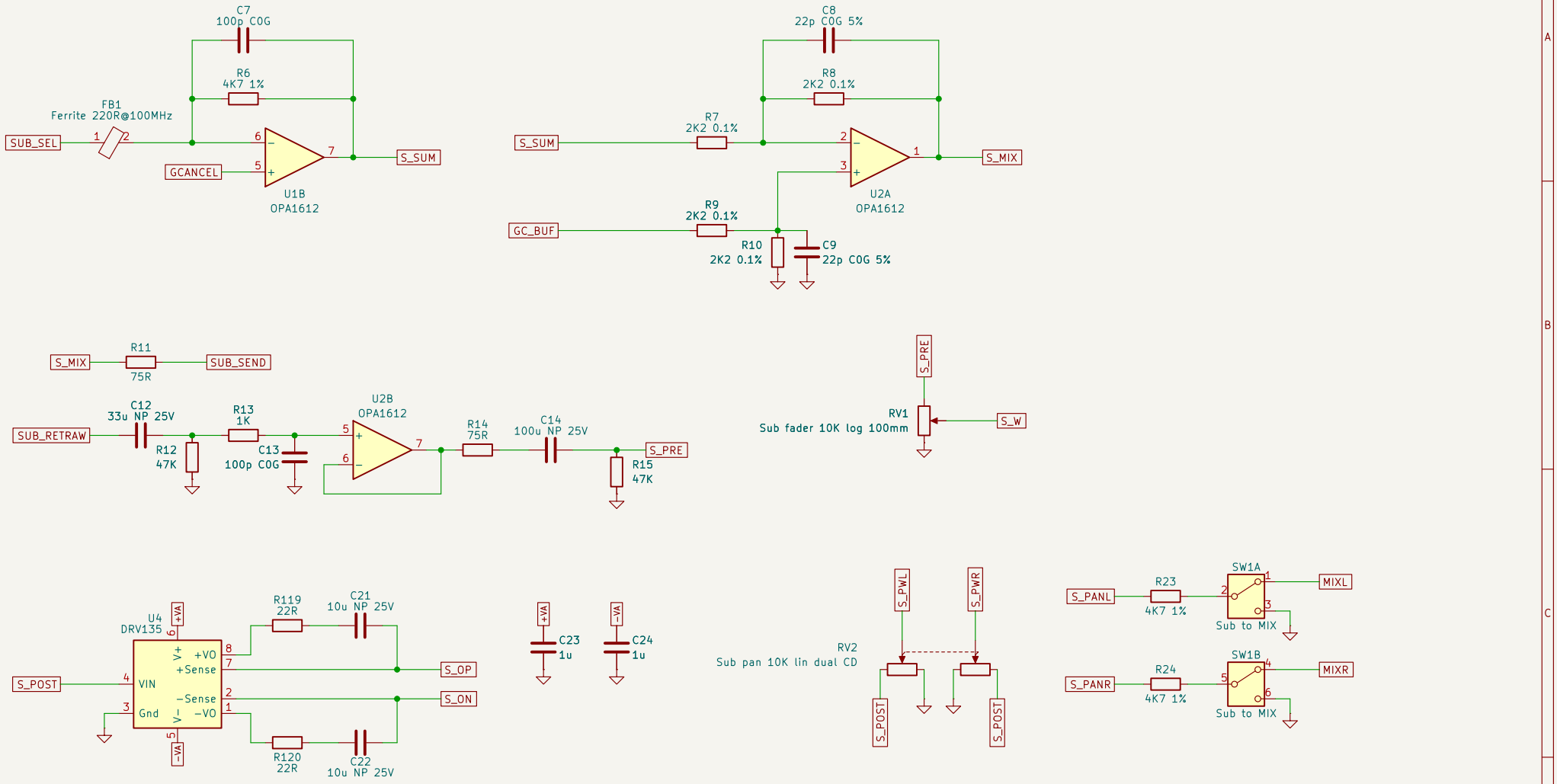
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /02 BUS SELECT, GROUND CANCEL BUFFER, SUBGROUP/
File: 02_bus_select.kicad_sch

Title: BUS SELECT, GROUND CANCEL BUFFER, SUBGROUP

Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 2/13

SUBGROUP: sum (selected SUB bus), re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL (cont.)



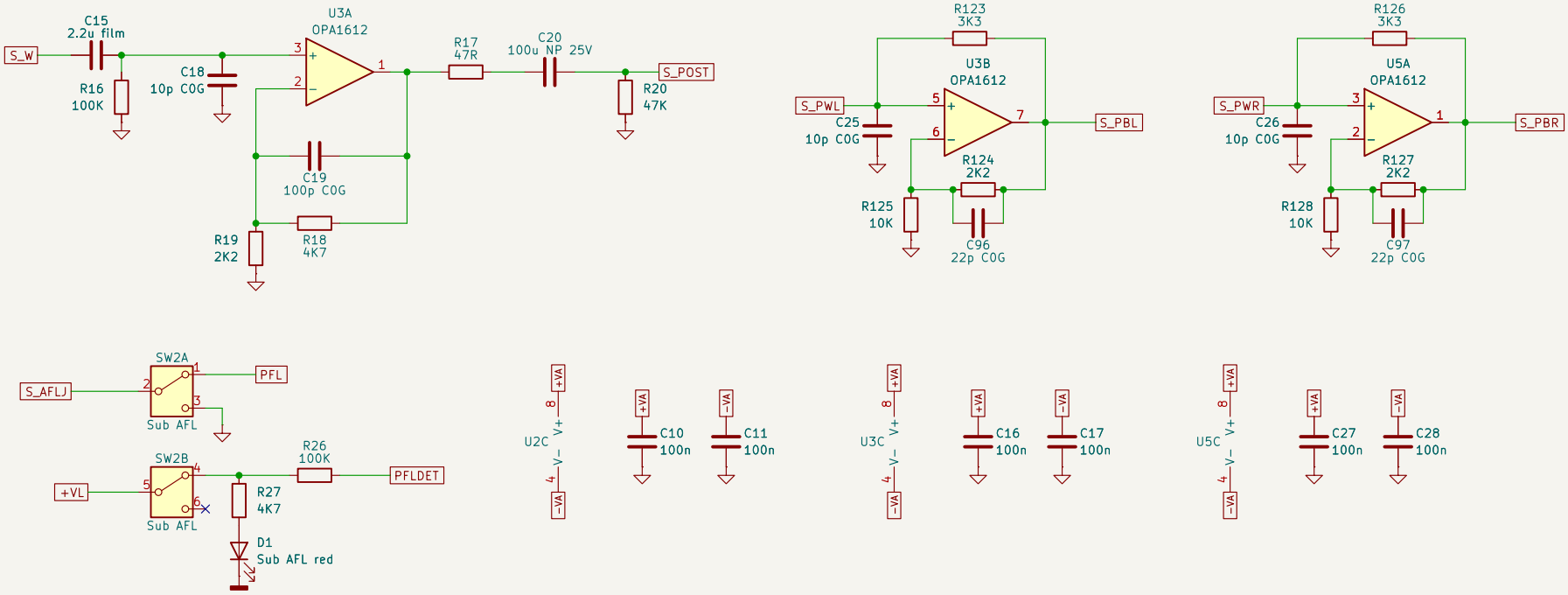
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /03 SUBGROUP/
File: 03_subgroup.kicad_sch

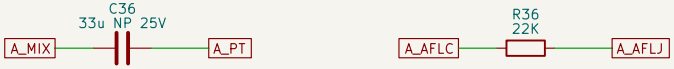
Title: SUBGROUP

Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 3/13

SUBGROUP: sum (selected SUB bus), re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL (cont.)



AUX MASTER: OPA1612 sum (selected AUX bus, 22K || 100p) + re-invert, level, buffer, DRV134 out, AFL [spec 13]



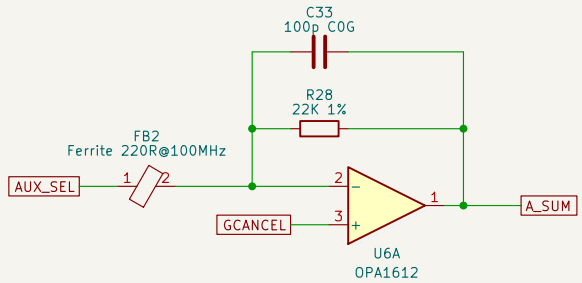
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

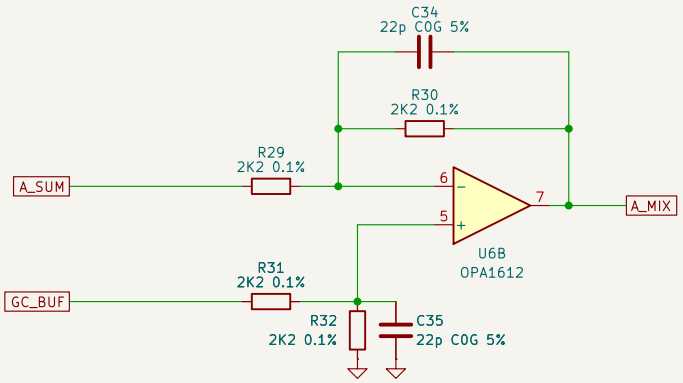
Sheet: /04 SUBGROUP, AUX MASTER/
File: 04_subgroup.kicad_sch

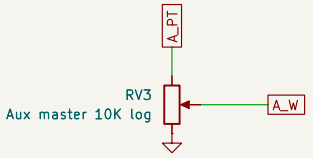
Title: SUBGROUP, AUX MASTER

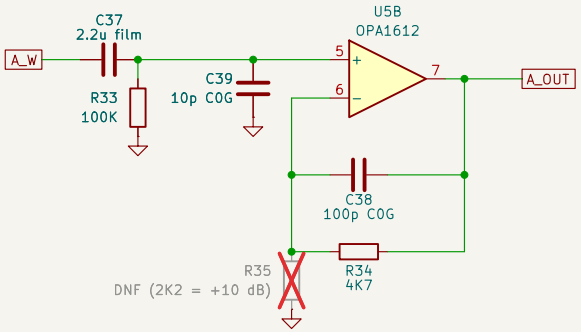
Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 4/13

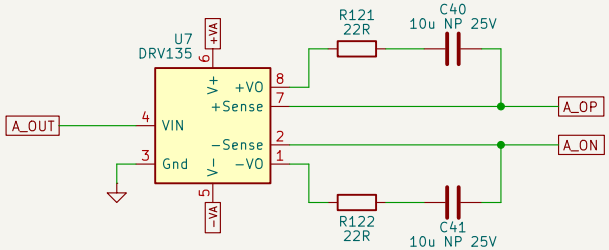
AUX MASTER: OPA1612 sum (selected AUX bus, 22K || 100p) + re-invert, level, buffer, DRV134 out, AFL (cont.)

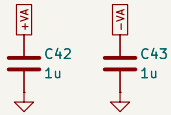


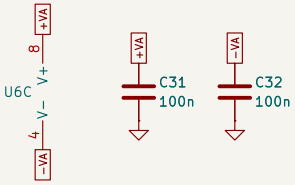














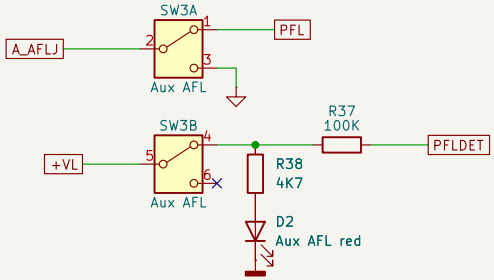
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /05 AUX MASTER/
File: 05_aux_master.kicad_sch

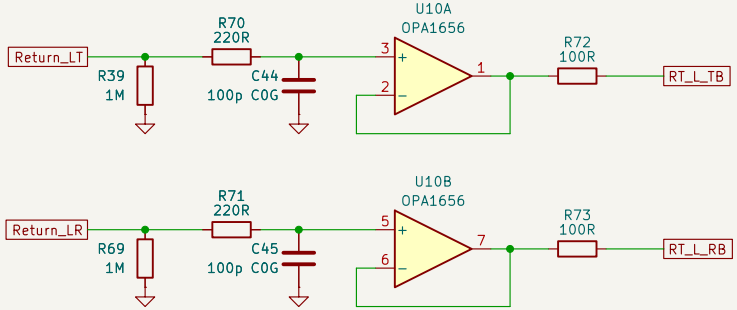
Title: AUX MASTER

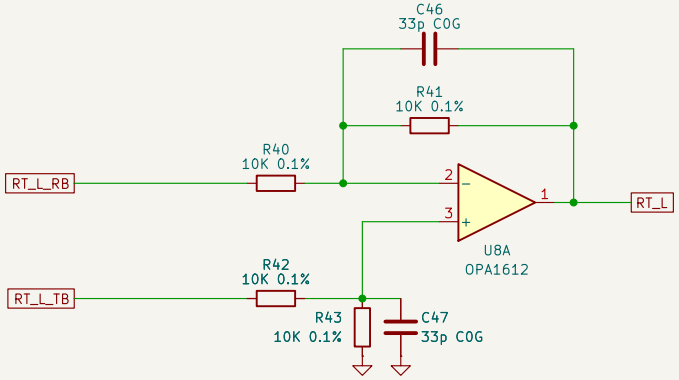
Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 5/13

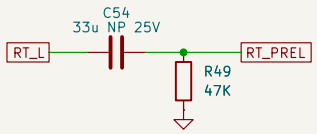
AUX MASTER: OPA1612 sum (selected AUX bus, 22K || 100p) + re-invert, level, buffer, DRV134 out, AFL (cont.)

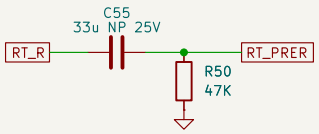


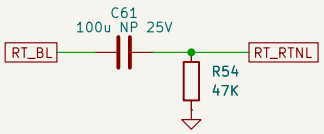
STEREO RETURN: receivers (R normalled to L), level, buffers, MIX / 1-2 / 3-4 onto the ribbon, PFL [spec 16]

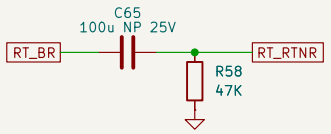












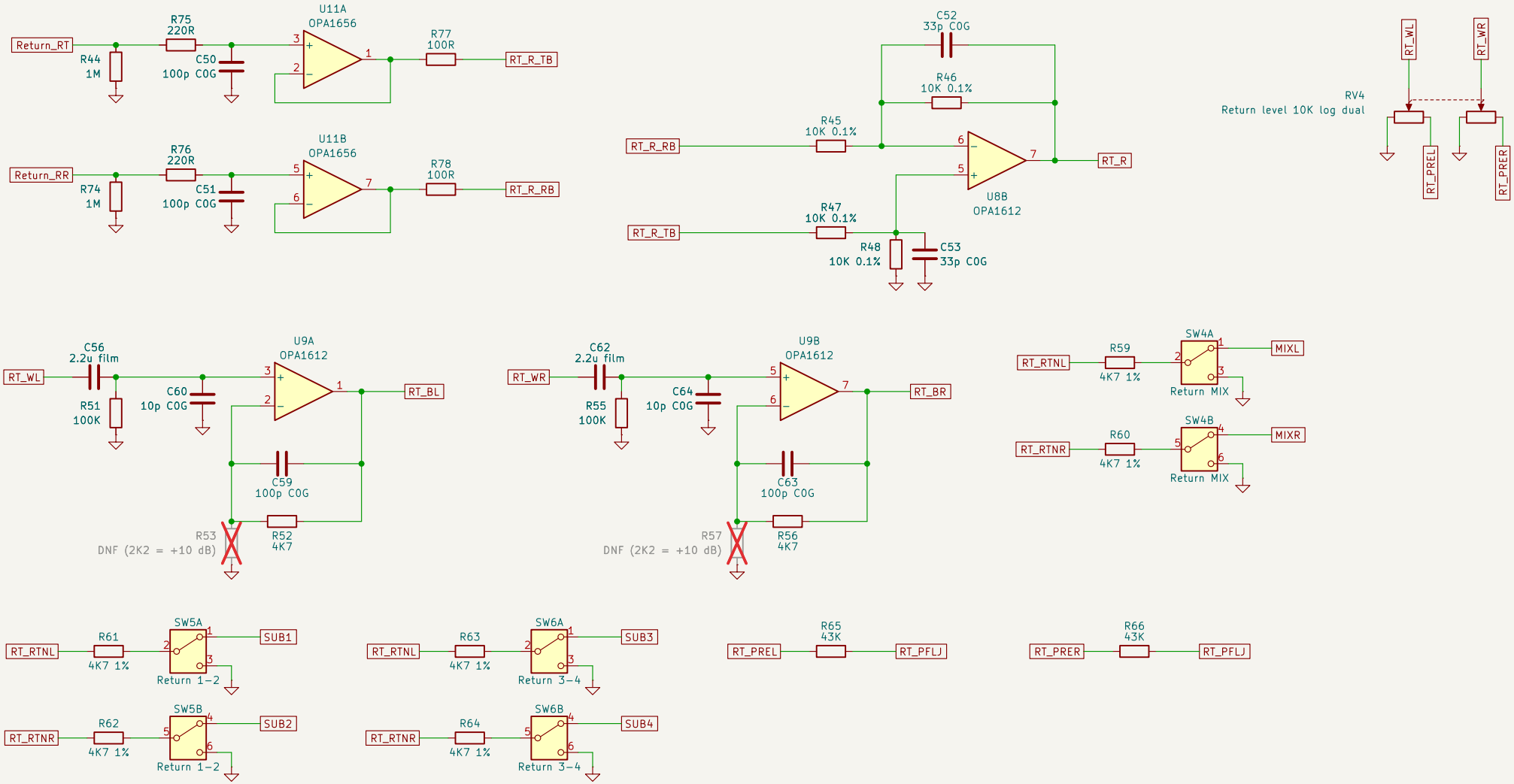
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /06 AUX MASTER, STEREO RETURN/
File: 06_aux_master.kicad_sch

Title: AUX MASTER, STEREO RETURN

Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 6/13

STEREO RETURN: receivers (R normalled to L), level, buffers, MIX / 1–2 / 3–4 onto the ribbon, PFL (cont.)



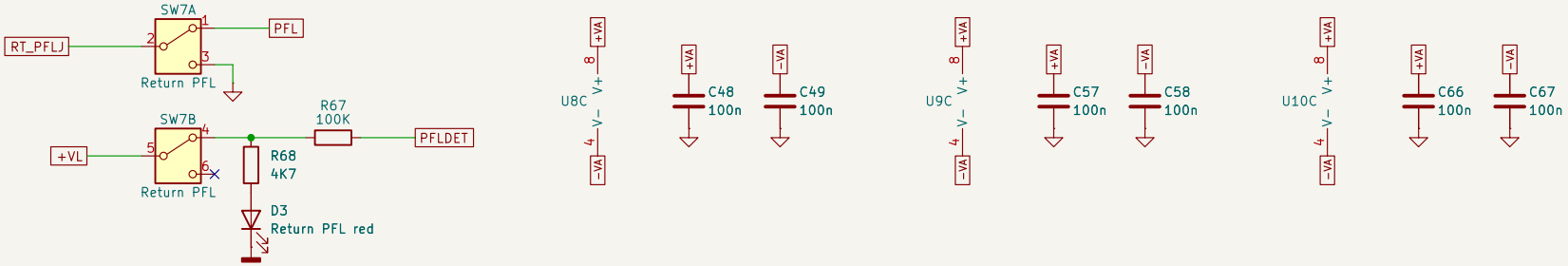
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

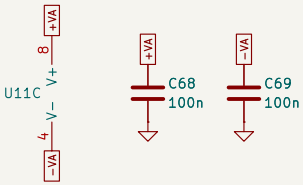
Sheet: /07 STEREO RETURN/
File: 07_stereo_return.kicad_sch

Title: STEREO RETURN

Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 7/13

STEREO RETURN: receivers (R normalled to L), level, buffers, MIX / 1–2 / 3–4 onto the ribbon, PFL (cont.)





STEREO RETURN 2: as return 1 (receivers, level, buffers, MIX / 1–2 / 3–4, PFL) [spec 16]



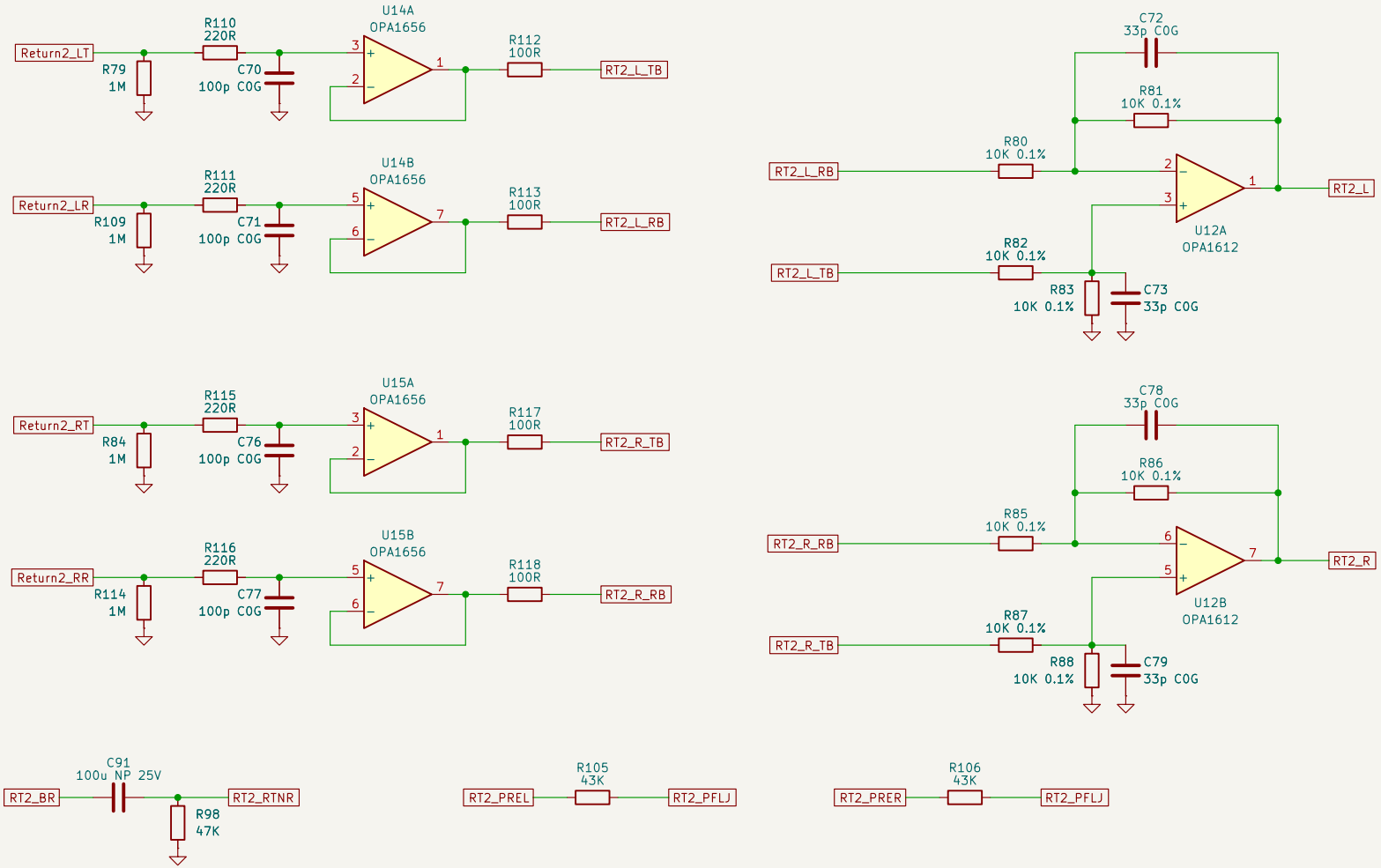
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /08 STEREO RETURN, STEREO RETURN 2/
File: 08_stereo_return.kicad_sch

Title: STEREO RETURN, STEREO RETURN 2

Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 8/13

STEREO RETURN 2: as return 1 (receivers, level, buffers, MIX / 1–2 / 3–4, PFL) (cont.)



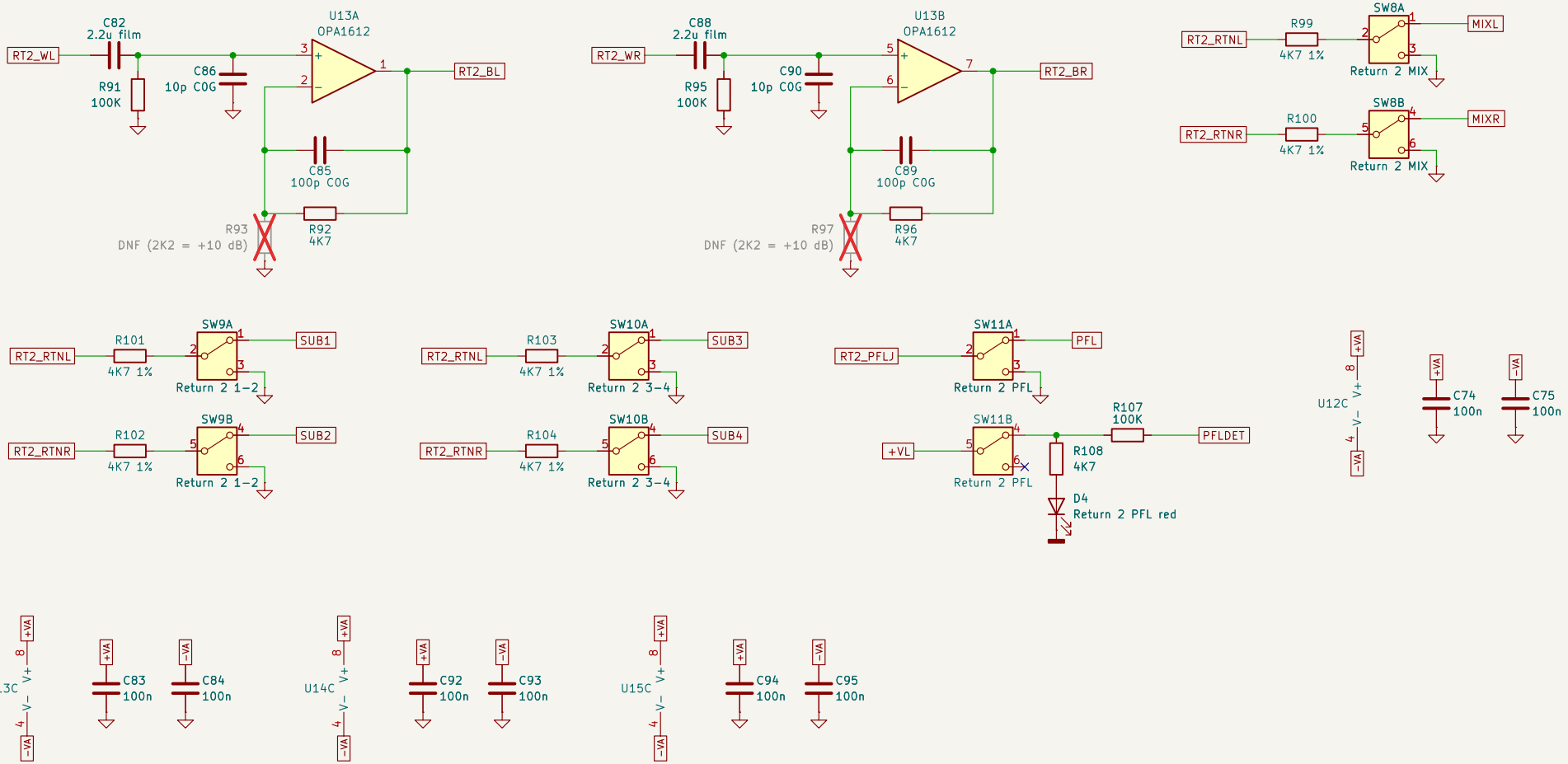
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /09 STEREO RETURN 2/
File: 09_stereo_return_2.kicad_sch

Title: STEREO RETURN 2

Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 9/13

STEREO RETURN 2: as return 1 (receivers, level, buffers, MIX / 1–2 / 3–4, PFL) (cont.)



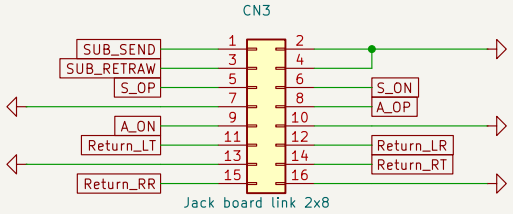
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module – goa console (sub + return + aux master)

Sheet: /10 STEREO RETURN 2/
File: 10_stereo_return_2.kicad_sch

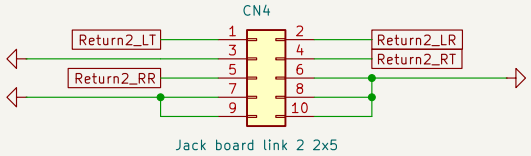
Title: STEREO RETURN 2

Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 10/13

JACK BOARD LINK 2x8 IDC ribbon to the group jack board (sub, insert, aux, return 1); bottom side, as the strip's CN3



JACK BOARD LINK 2 2x5 IDC ribbon to the group jack board (return 2); bottom side, beside the first



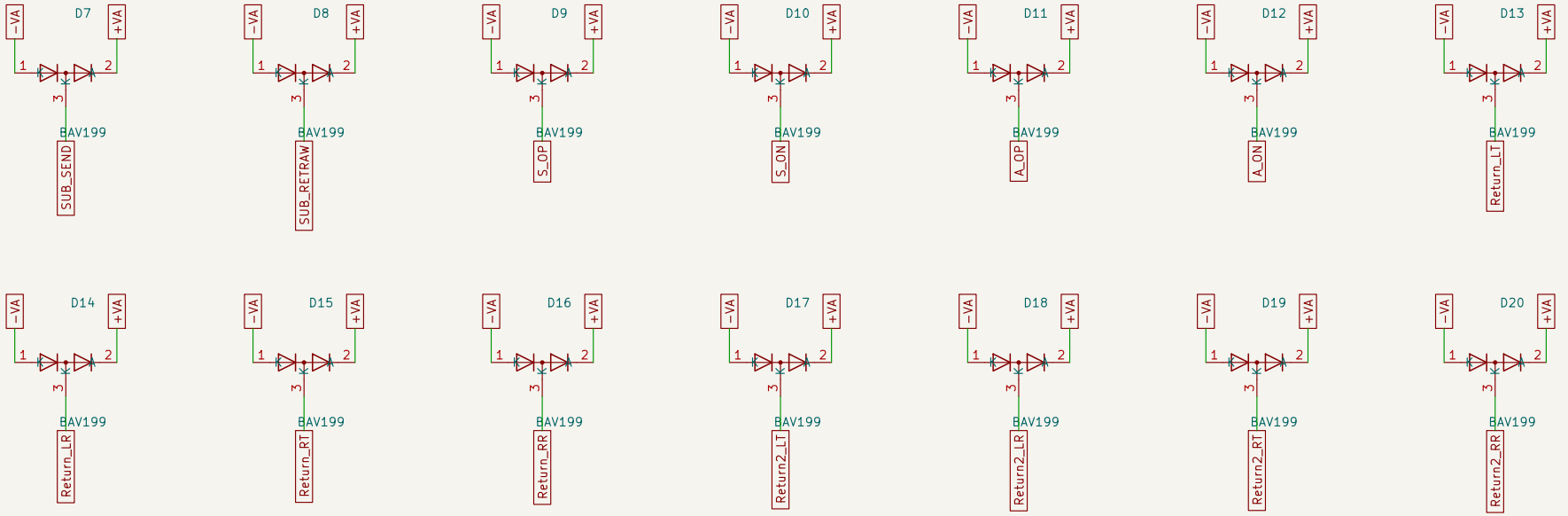
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /11 JACK BOARD LINK, JACK BOARD LINK 2/
File: 11_jack_board_link.kicad_sch

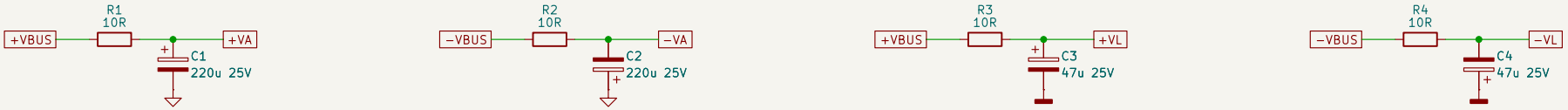
Title: JACK BOARD LINK, JACK BOARD LINK 2

Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 11/13

JACK LINK CLAMPS: a BAV199 to the audio rails on every jack signal at the links (CS4b)



POWER AND GROUNDS (as a strip: 10R + 220u per audio rail, 10R + 47u logic rails, spade, 10R ground cancel)



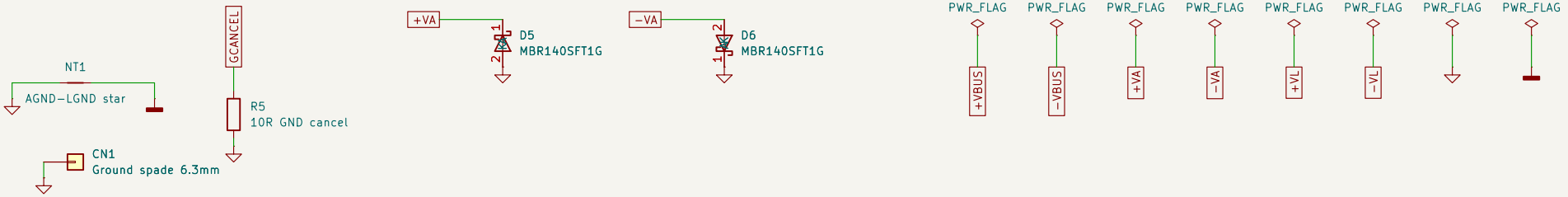
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /12 JACK LINK CLAMPS, POWER AND GROUNDS/
File: 12_jack_link_clamps.kicad_sch

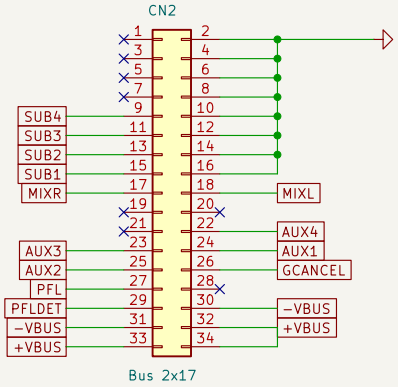
Title: JACK LINK CLAMPS, POWER AND GROUNDS

Size: A4	Date: 2026-10-19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 12/13

POWER AND GROUNDS (cont.)



RIBBON one 34–pin header, Delta pinout (as a strip); AUX 5–6 are summed on the master module



Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module – goa console (sub + return + aux master)

Sheet: /13 POWER AND GROUNDS, RIBBON/
File: 13_power_and_grounds.kicad_sch

Title: POWER AND GROUNDS, RIBBON

Size: A4	Date: 2026–10–19	**Rev: 0.12**
KiCad E.D.A. 10.0.7		Id: 13/13